

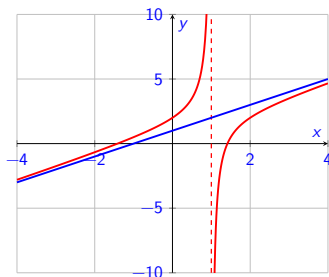
2627-MA140: Week 02, Lecture 2 (L05)

Introduction to Limits

Dr Niall Madden

University of Galway

Wednesday, 16 September, 2026



Slides by Niall Madden, with some material adapted from textbooks, and original notes by Dr Kirsten Pfeiffer.

Today's class is limited to these topics:

- 1 Today's class is limited to these topics:
- 2 Reminders
- 3 Recall: towards "limits"
- 4 Definition of a Limit
- 5 Properties of Limits
 - Evaluating limits
- 6 Limits of rational functions
- 7 Completing the square
- 8 Exercises

For more, see Chapter 2 (Limits) of Strang and Herman's **Calculus**, especially Sections 2.2 (Limit of a Function) and 2.3 (Limit Laws).

Slides are on canvas, and at niallmadden.ie/2627-MA140



Reminders

- ▶ Tutorials started **this** week.
- ▶ Current assignment (for this week's tutorials) is **PS-0**. Just for practice. See Canvas for link.
- ▶ **Assignment 1** (**PS-1**) opens tomorrow, and due 5pm, Monday 28 October. Will be covered in tutorials next week.

Recall: towards "limits" neither f nor g have $x=1$ in their domains.

Yesterday, we considered the following functions:

$$f(x) = \frac{x^2 - 2}{x - 1} \qquad g(x) = \frac{x^2 - 1}{x - 1}$$

We noticed that, although neither f nor g is defined at $x=1$, they behaved very differently as x approaches 1.

To discuss this we'll need the concept of a **limit** which, roughly, relates to the value of function as it **approaches** a point (but not actually at that point).

$$f(1) = -\frac{1}{0} \qquad g = \frac{0}{0} ??$$

$$\lim_{x \rightarrow a} f(x) = L$$

{ The concept of a limit is a prerequisite for a proper understanding of calculus, and numerical methods. }

Definition of a Limit

$\varepsilon \approx$ "epsilon"

Some conventions and terminology we'll use:

- ▶ x is a variable.
- ▶ a is a fixed number: we are interested in values of x near a .
- ▶ ϵ is a **small** positive number; we can choose it as small as we like.
- ▶ δ is another **small** positive number; it is determined by ϵ .
- ▶ $|x - a| < \delta$ means that the distance between x and a is less than δ .
 $\delta =$ "delta"
 $\text{ie } x - \delta < a < x + \delta$
- ▶ As x approaches a , so $f(x)$ approaches a number L .

When we write:

$$\lim_{x \rightarrow a} f(x) = L,$$

we read:

"The limit of f , as x goes to a , is L ".

Definition of a Limit

LIMIT: formal definition

$$\lim_{x \rightarrow a} f(x) = L,$$

means that, for every number $\epsilon > 0$, it is possible to find a number $\delta > 0$, such that

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad |x - a| < \delta.$$

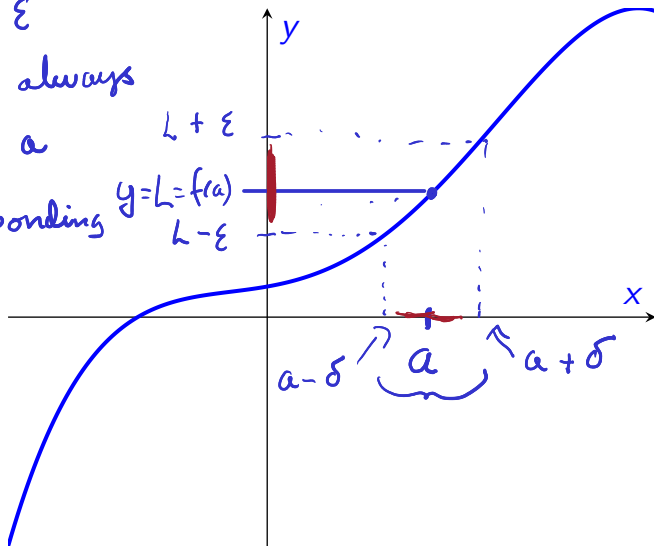
LIMIT: Informal explanation

$$\lim_{x \rightarrow a} f(x) = L,$$

means that we can make $f(x)$ as close to L as we like, by taking x as close to a as needed.

Definition of a Limit

Given ε
I can always
find a
corresponding
 δ .



Definition of a Limit

Example

Prove formally that $\lim_{x \rightarrow 3} (4x - 5) = 7$.

That is, for arbitrary ϵ , find a δ such that

$$|(4x - 5) - 7| < \epsilon \quad \text{whenever} \quad |x - 3| < \delta.$$

Here $f(x) = 4x - 5$ & $L = 7$.

We want $|(4x - 5) - 7| < \epsilon$

$$\Rightarrow |4x - 12| < \epsilon.$$

$$\Rightarrow 4|x - 3| < \epsilon \Rightarrow |x - 3| \leq \frac{\epsilon}{4}.$$

So take δ to be any number less than $\frac{\epsilon}{4}$.
Eg, $\delta = \frac{\epsilon}{5}$.

Definition of a Limit

The approach we just used is technically correct, but not very practical in many cases.

Fortunately, there are other methods that can be used

- ▶ to show that a limit exists;
- ▶ find the limit of a function, $f(x)$ as $x \rightarrow a$.

Properties of Limits

See also...

... Section 2.3 of the textbook: **Limit Laws**

Suppose that $\lim_{x \rightarrow a} f_1(x) = L_1$, and $\lim_{x \rightarrow a} f_2(x) = L_2$ and $c \in \mathbb{R}$ is any constant. Then,

(1) $\lim_{x \rightarrow a} c = c$, This is, if $f(x) = c$ for all x (ie the graph of f is a horizontal line!) then

$$\lim_{x \rightarrow a} f(x) = c.$$

(2) $\lim_{x \rightarrow a} x = a$

That is, if $f(x) = x$ then
$$\lim_{x \rightarrow a} f(x) = f(a) = a.$$

Properties of Limits

$$(3) \lim_{x \rightarrow a} [cf_1(x)] = cL_1$$

$$\text{eg } \lim_{x \rightarrow a} (5x) = 5 \lim_{x \rightarrow a} x = 5(a) = 5a.$$

$$(4) \lim_{x \rightarrow a} [f_1(x) + f_2(x)] = L_1 + L_2 \text{ and}$$

$$\lim_{x \rightarrow a} [f_1(x) - f_2(x)] = L_1 - L_2$$

$$\lim_{x \rightarrow a} (f_1(x) + f_2(x)) = \lim_{x \rightarrow a} f_1(x) + \lim_{x \rightarrow a} f_2(x).$$

$$\begin{aligned} \text{eg, } \lim_{x \rightarrow 3} (4x - 5) &= \lim_{x \rightarrow 3} (4x) + \lim_{x \rightarrow 3} (-5) \\ &= 4(3) - 5 = 7. \quad (\text{By laws 1, 2 \& 3}) \end{aligned}$$

Properties of Limits

$$(5) \lim_{x \rightarrow a} (f_1(x) f_2(x)) = L_1 L_2$$

$$\lim_{x \rightarrow a} (f_1(x) f_2(x)) = \lim_{x \rightarrow a} f_1(x) \lim_{x \rightarrow a} f_2(x).$$

$$\text{Eg } \lim_{x \rightarrow 3} (x^2) = \left(\lim_{x \rightarrow 3} x \right) \left(\lim_{x \rightarrow 3} x \right) = (3)(3) = 9.$$

$$(6) \lim_{x \rightarrow a} ((f_1(x))^n) = (L_1)^n, \text{ for any positive integer } n.$$

$$\text{Eg } \lim_{x \rightarrow a} (x^3) = \lim_{x \rightarrow a} (x) \lim_{x \rightarrow a} (x^2) = a^3$$

in general

$$\lim_{x \rightarrow a} ((f_1(x))^n) = \lim_{x \rightarrow a} \underbrace{f_1(x) f_1(x) \cdots f_1(x)}_{n \text{ times}} = (L_1)^n$$

Properties of Limits

$$(7) \lim_{x \rightarrow a} \left(\frac{f_1(x)}{f_2(x)} \right) = \frac{L_1}{L_2}, \quad \text{providing } L_2 \neq 0.$$

$$\lim_{x \rightarrow a} \left[\frac{f_1(x)}{f_2(x)} \right] = \frac{\lim_{x \rightarrow a} f_1(x)}{\lim_{x \rightarrow a} f_2(x)} \quad \leftarrow \text{not zero!}$$

$$(8) \lim_{x \rightarrow a} \sqrt[n]{f_1(x)} = \sqrt[n]{L_1} \\ = \left(\lim_{x \rightarrow a} f_1(x) \right)^{1/n}.$$

Note: we can combine these properties as needed. For example, (5) and (8) together give that

$$\lim_{x \rightarrow a} x^n = a^n$$

Example

Evaluate the limit $\lim_{x \rightarrow 1} (x^3 + 4x^2 - 3)$

By Law 4:

$$\lim_{x \rightarrow 1} (x^3 + 4x^2 - 3) = \lim_{x \rightarrow 1} (x^3) + \lim_{x \rightarrow 1} (4x^2) + \lim_{x \rightarrow 1} (-3)$$

$$= 1^3 + 4 - 3$$

Law 6 Law 3 & 6 Law 1

Example

Evaluate $\lim_{x \rightarrow 1} \frac{x^4 + x^2 - 1}{x^2 + 5}$ using the Properties of Limits.

$$\lim_{x \rightarrow 1} \frac{x^4 + x^2 - 1}{x^2 + 5} = \frac{\lim_{x \rightarrow 1} (x^4 + x^2 - 1)}{\lim_{x \rightarrow 1} (x^2 + 5)}$$

$$= \frac{1 + 1 - 1}{1 + 5} = \frac{1}{6} \quad \checkmark$$

$\neq 0$

Here we applied Law 7 since $L_2 \neq 0$.

Limits of rational functions

In many cases, evaluating limits is more complicated. In particular, we'll consider numerous examples where we want to evaluate $\lim_{x \rightarrow a} f(x)$ where a is not in the domain of f .

A typical example of this is when we evaluate a rational function:

$$\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$$

where **both** $p(a) = 0$ and $q(a) = 0$. ✓

Idea: Since we care about the value of p and q **near** $x = a$, but not actually at $x = a$, it is safe to factor out an $(x - a)$ term from both. (since both are polynomials).

Limits of rational functions

Three examples

Evaluate the limits:

$$(a) \lim_{x \rightarrow 0} \frac{x}{x} \quad (b) \lim_{x \rightarrow 0} \frac{x^2}{x} \quad (c) \lim_{x \rightarrow 0} \frac{x}{x^2}$$

$$(b) \lim_{x \rightarrow 0} \frac{\cancel{x}(x)}{\cancel{x}} = \lim_{x \rightarrow 0} x = 0 \quad \checkmark$$

$$(c) \lim_{x \rightarrow 0} \frac{\cancel{x}}{\cancel{x}(x)} = \lim_{x \rightarrow 0} \frac{1}{x} \quad \text{which is not defined.}$$

Finished here Wednesday

Limits of rational functions

Three examples

Evaluate the limits:

$$(a) \lim_{x \rightarrow 0} \frac{x}{x} \quad (b) \lim_{x \rightarrow 0} \frac{x^2}{x} \quad (c) \lim_{x \rightarrow 0} \frac{x}{x^2}$$

(a) $p(x) = x$ $q(x) = x$. So, if $x = 0$, $p(x) = q(0) = 0$

But $\lim_{x \rightarrow 0} \frac{x}{x} = \lim_{x \rightarrow 0} \frac{1}{1}$ Since $x \neq 0$

$$\text{So } \lim_{x \rightarrow 0} \frac{x}{x} = 1$$

Limits of rational functions

Example

Evaluate the limit

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$$

Limits of rational functions

In that last example, we found that

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x + 2}{x}$$

But these are different functions:

Limits of rational functions

Evaluate the limit

$$\lim_{x \rightarrow 2} \left(\frac{\frac{1}{2} - \frac{1}{x}}{x - 2} \right)$$

Completing the square

Very often, we'll evaluate limits of the form:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where f and g are not polynomials. Some of the same ideas still apply.

Example

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - 1}{x^2}$$

Completing the square

Exercise 2.2.1

Evaluate the following limits

$$(a) \lim_{x \rightarrow \frac{1}{2}} \frac{x - \frac{1}{2}}{x^2 - \frac{1}{4}}$$

$$(b) \lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x^2 + x - 12}$$

Exercise 2.2.2

(From 2023/2024 MA140 exam, Q1(a)) Evaluate the limit

$$\lim_{x \rightarrow 4} \frac{x - 4}{(\sqrt{x} - 2)(x + 9)}$$

Exercises

Exercise 2.2.3 (from 2526-MA140 exam)

Let $f(x) = \frac{x^2 - x - 12}{x^2 - 7x + 12}$. For each of the following, evaluate the limit, or explain why it does not exist.

1. $\lim_{x \rightarrow -4} f(x)$
2. $\lim_{x \rightarrow 4} f(x)$
3. $\lim_{x \rightarrow \infty} f(x)$